

Empowering Students as Leaders of Co-Design for Block-Based Programming

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ABSTRACT

Educational software is rarely co-designed with teachers and students. We have created a novel co-design process that can help both teachers and students learn about human-computer interaction (HCI) and research methods while contributing to the design of educational systems that will benefit future students and teachers. We investigated the results of our process of mentoring two undergraduate students to lead participatory design (PD) sessions with three K-12 teachers. We found that this process exposed teachers and undergraduates to HCI research and design techniques, empowered undergraduates to lead design sessions, and encouraged knowledge transfer between the groups. Our lessons learned include ways to prepare students inexperienced with HCI to lead PD, with training in the participants' domain, practice sessions, and predefined PD methods to adapt for their context. Our recommendations for future design collaborations can also provide insights to empower undergraduate students or less experienced teams and participants to benefit from participatory design efforts.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**.

KEYWORDS

Participatory Design, K-12 computing, Undergraduate Research, HCI Teaching Methods

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1 INTRODUCTION

Educational software is rarely co-designed with teachers and students. Instead, teams of developers and instructional designers often

create software that targets improved learning for individual students, but fails to address the needs of teachers in each stage of their teaching process, which is a major barrier to adoption of such systems in classrooms. We have created a novel co-design process that can help both teachers and students learn about human-computer interaction (HCI) and research methods while contributing to the design of educational systems that will benefit future students and teachers.

Teachers need software for their classrooms that is designed to fit their workflow and needs. Past co-design has shown a range of considerations that must go into learning technologies in order to support teachers, including persevering their autonomy as teachers [2], supporting different classroom formats [28], maintenance of hardware [27] and mappings of learning environment content to curricula [28]. While past findings can help guide the development of new learning software, needs may vary based on the domain, grade level, and cultural context.

While co-design is necessary to understand the unique context and needs of users, it unfortunately can be challenging for students and research labs at universities to hire HCI experts to help with the design of learning software. This means that sometimes no user studies are conducted on university produced software. Enabling students to conduct co-design sessions - although imperfect - may result in better software creation.

On the other hand, there is a need for computer science (CS) undergraduates to have practical educational experiences that prepare them for the workforce. Outside of internships and one-semester capstone courses, few CS undergraduates gain design experience working with clients on legacy projects. HCI courses often focus on designing from scratch, rather than extending existing (legacy) systems or engaging in popular industry HCI methods. Co-design experiences for undergraduates could also greatly benefit users of software. Research has shown that the co-design process has been deployed successfully in educational settings with teachers benefiting from the creative and technical effort of students making computing lessons [16]. Similarly, it is possible that teachers could benefit from software that comes out of a co-design process with students as it may better meet their needs.

Our institution has a unique environment where undergraduate students and in-service teachers engage in full-time summer research experiences to learn about research and computing [5]. Additionally, our research lab creates educational software. So, we designed a summer program to uniquely meet the needs of the student and teacher researchers while also contributing to improving our educational software through a co-design process.

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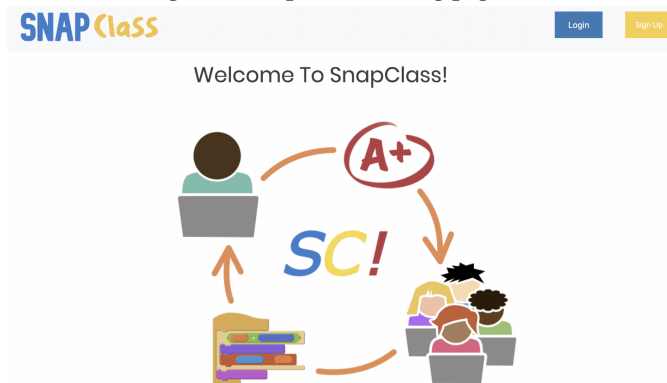
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We mentored two Research Experience for Undergraduate students (REUs), to organize and lead participatory design sessions with three teachers (for grades 6–12) in the summer program. Under the guidance of two research mentors, the REUs read related literature, created protocols, and facilitated design sessions with the teachers. The goal of the sessions were to ideate and design new features for SnapClass, a web platform for the block-based programming language, Snap! [14]. In SnapClass, students can complete their programming assignments and teachers can assign, define rubrics, and grade with their rubrics (Figure 1). SnapClass was an optimal option for the co-design as the teachers will likely use the software in their classes in Fall 2023. The outputs of the co-design process were features, requirements, and specifications for future graduate and undergraduate computer science majors to implement.

Through thematic analysis [22] we investigate the outcomes of the design collaboration and provide recommendations for future design collaborations between REUs and teachers. Specifically, we address the following questions: **RQ1**: To what extent was this methodology successful at empowering undergraduate students (REUs) to lead a participatory design session with computer science teachers of grades 6–12? **RQ2**: To what extent was our methodology successful at incorporating teachers (RETS) into the design process? **RQ3**: To what extent was our methodology successful at creating useful and implementable designs?

Through addressing these research questions in our discussion, we hope to improve future implementations of this program and provide insights for other university mentors to empower undergraduate students to lead participatory design at their own institutions. These insights might further be useful for small teams to bootstrap HCI efforts with employees and clients unfamiliar with participatory design methods.

Figure 1: SnapClass landing page.



2 RELATED WORK

It's been shown, through a multiple case study, that for classroom innovation to be sustainable, it is important to have both teacher and student support of the innovation [27]. A learning environment's effectiveness relies on the student perceptions of that environment [18]. Therefore, it is evident that teachers and students should

participate in the design process of learning technologies to ensure any innovation will be well adopted. Fortunately, past research has shown that both students and teachers at a secondary level feel positively about participating in the participatory design (PD) process [19]. And, there is already a long established history of using PD for educational software as demonstrated by recent literature reviews [3, 40].

PD methods have been used to design learning technologies in many domains including science [13, 23, 37], mathematics [12, 38], music [11], braille writing [17], literature [29], data visualization for social studies [25, 26], and collaborative learning [32]. In the computing education space, PD with teachers has been used to create culturally relevant computing curriculum [7] and a User Experience sprint was conducted to design a block-based grading system [24]. To our knowledge this is the first study to use the Make-Tell-Enact PD process [4, 34] with teachers to design a learning technology to support programming. *Telling* activities are those that enable participants to share their experiences in, and knowledge of, their domain [4]. *Making* activities include early prototyping with participants or having participants create hypothetical representations of their futures [4, 34]. Finally, *Enacting* prompts participants to explore how new designs might affect their and other users' workflows, behaviors, and experiences [4].

A few PD studies with the goal of creating learning technologies have been conducted with students as participants [15, 31, 35, 36]. Some students have been taught how to facilitate PD in university courses [6, 8]. However, to our knowledge, no studies have empowered students to facilitate PD sessions in a real research setting. As such, there is a demand in the research literature for mechanisms to best incorporate students learning HCI methods in leading PD which is the focus of our study. Similarly, many user experience (UX) and HCI positions at small companies are filled by practitioners with low prior experience [9]. And, there is no one standard degree or program for professionals in the field of UX [30]. Understanding how to support low prior experience facilitators, like our students, is crucial. Our undergraduate REU students not only learn about PD, but they design their own PD sessions and facilitate the sessions with teachers.

3 METHODS

3.1 Context - Undergraduate REU & Teacher Summer Programs

Two undergraduate students participated in our 10-week summer computing Research Experiences for Undergraduates (REU) program. Upon arrival, REUs are introduced to possible projects to ensure that students are participating and gaining skills in a project of interest to them. These students (a rising 2nd year and a rising 3rd year, both female) chose our research team indicating they were motivated and excited to work on a design project with teachers (of grades 6–12) where they would learn HCI methods. During their first two weeks in the program, the REUs - with assistance from a graduate mentor - read literature related to PD. Using the literature, the REUs chose PD methods and created plans for carrying out each method. They then ran practice sessions with graduate and postdoc mentors to receive feedback on their design. The methods that the REUs chose are detailed below. During the remaining seven weeks

of the program, the undergraduate researchers were tasked with implementing the co-designed features into the existing SnapClass platform. In the remainder of this paper, we refer to the two undergraduate students facilitating co-design sessions as REUs; we use the word “student” when we refer to their inputs in the design process from a learner perspective.

During the same summer, we invited 3 in-service teachers to participate in a full-time, 6-week research and curriculum development program. Two of the teachers (1M, 1F) taught science at a magnet middle school where computational thinking is integrated into several subjects using programming lessons in Snap!. One teacher (F) taught introductory programming at a public high school using Java and Scratch. Our novel program was designed to engage both undergraduates and teachers in research in computing education, HCI, and curriculum design.

3.2 REU-Chosen Methods for engaging Teachers in PD

Inspired by the Make, Tell, Enact cycle [4, 33, 34], REUs planned and led design sessions. The PD sessions took place over the course of four days. The session audio was recorded and the collected design artifacts include drawings, notes, and designs created in Figma [10], a low-fidelity prototyping software. The progression of the sessions is outlined in Figure 2. The REUs and teachers worked in a secluded conference room with markers, paper, sticky notes, scissors, and tape at the table. The room also had a monitor so the participants could share their computer screens.

3.2.1 Tell: Future Workshop. The future workshop [20, 39] consisted of four phases: Brainstorming, Critical Appraisal, Utopian Fantasy, and Implementation. Throughout each session, the REUs asked the teachers to draw, write, and construct with the materials provided to them. 1) Brainstorming: teachers were instructed to describe their current process of assigning and grading programming assignments and outline their typical workflow. 2) Critical Appraisal: teachers identified their pain points within their grading and activity assignment processes by first reflecting individually and then sharing with the group. 3) Utopian Fantasy: Teachers described a world in which their problem would no longer exist, often describing a technology solution designed to fix the problem. Teachers were instructed to think beyond what they believed current technology was capable of. 4) Implementation: During the workshop, teachers identified more realistic software solutions that may help them get closer to their utopias. The REUs decided to introduce the existing SnapClass tool after the future workshop so that teachers would not be limited by their perception of what they thought the software was capable of and so they focus on larger features to add rather than small bug fixes.

3.2.2 Make Session. After introducing the teachers to SnapClass, the REUs led a ‘Make’ session combining some elements of a Sketch-in [21] with a Live Prototyping session. First, teachers were asked if the current version of SnapClass solved any of their pain points identified in the future workshop. Additionally, they were asked to identify their overall impressions of SnapClass. Then, using their pain points and utopian solutions, the teachers created a list of solutions that may be useful to integrate into the software. Each

solution was discussed and elaborated upon then the teachers each chose a solution that was highest priority to them. After the REUs introduced Figma and taught the teachers the basics, each teacher either drew their idea on paper in more detail or using an existing Figma template, which had some of the SnapClass functionality, to model their idea. Each teacher then shared what they designed and received input from the other teachers. The REUs cleaned and refined the teachers’ designs with the feedback given.

3.2.3 Enact. The purpose of an enact session is to review and test designs from the perspectives of the users - teachers and students - in the range of activities they may use the tool. Instead of using a specific predefined enact method, the REUs created an protocol inspired by related literature. Their protocol is as follows: After refining the designs, the REUs met with the teachers. The teachers were asked questions and given feedback about the designs from their own and student user perspectives. Additionally, they were prompted to use the Figma prototypes and act out how they would use the prototype in their classrooms. The REUs then took feedback from teachers to refine their designs.

3.3 Data Collection & Analysis

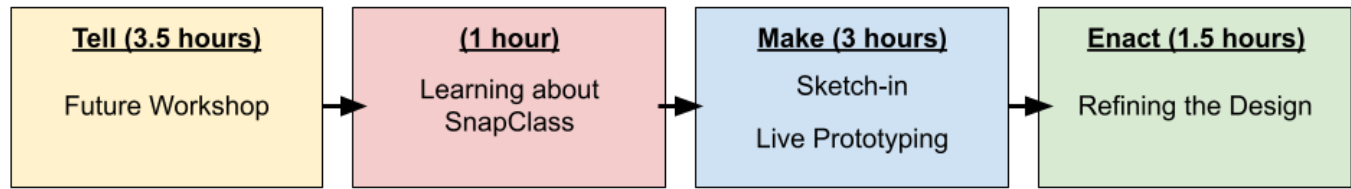
Two researchers, who were present for a subset of the sessions and were familiar with SnapClass, analyzed the session recordings. The goal was to evaluate the PD sessions in terms of how well the novice REU researchers were able to lead. The researchers watched the session recordings and performed a thematic analysis [22]. The focus of the thematic analysis was not on the teachers’ needs and designs, but instead on the program design experience. This includes the collaboration between REUs and teachers, attitudes towards the activities, project successes, project pain points, and the knowledge and skills gained by each group. After the researchers coded the session recordings separately, they met to discuss and agree upon the tags forming an initial set of 236 tags. Together the researchers sorted the tags; 39 themes were found, which were then sorted into five categories. These high-level categories are discussed below.

Analysis was focused on data that would help answer the following questions: **RQ1:** To what extent was this methodology successful at empowering undergraduate students (REUs) to lead a participatory design session with computer science teachers of grades 6-12? **RQ2:** To what extent was our methodology successful at incorporating teachers into the design process? **RQ3:** To what extent was our methodology successful at creating useful and implementable designs?

4 RESULTS

4.1 REUs as facilitators - RQ1

One strength of the undergraduate (REU) facilitators was their ability to ask questions to refine and clarify teacher designs. One REU asked about implementations details of a self-assessment rubric; “How is it different for every assignment? Or is it the same for every assignment?”. This helped the teacher imagine a broader range of use cases. Even though REUs asked good probing questions, they often did not know to what extent they should probe. When transitioning from critical appraisal to utopian fantasy phase, one REU

Figure 2: The sequence of activities REU and teacher participants engaged in.

asked the teachers and the research mentors, “So are we done with this stage? Or...?” showing their uncertainty about appropriate timing. Similarly, when receiving feedback on their designs, the REUs moved on to the next feature, before ending discussion on the previous feature. Demonstrating this, one teacher interjected: “one thing though, about the previous one...”. The REUs continued in this uncertain manner throughout the session. Oftentimes, the REUs would look to the mentors when uncertain how to run the session or answer a teacher question. One of the REUs asked the mentors how they should introduce the SnapClass interface to the teachers: “Should I also have it on [my laptop] and then just share it on the screen?”. Many times the mentors needed to explain further the tasks that the teachers were to be completing. A mentor needed to clarify the goals of a utopian brainstorming session, explain how SnapClass works, and step in to help guide the Figma teacher tutorial. The REUs also often asked teachers if they could work on something showing their uncertainty in their leadership role. When introducing the teachers to SnapClass, one REU said “If you guys don’t mind, I guess take turns working on it... Can you just take turns working on it from here?”.

4.2 Successful Collaboration between teachers and REUs - RQ1 & RQ 2

A unique discourse opportunity occurred between the teachers and REUs, as both were potential software users and both had an understanding of classroom dynamics. REU perspectives were valued by teachers. The teachers often asked REUs for design advice or for more information about their perspectives as students. For example, one teacher expressed frustration that usually a small subset of group members complete most of the work. Then, the teacher asked an REU about how when working in a research group, each person can take ownership of part of the project; “What are some of my options?”. The REUs were proactively providing their perspectives as students. When designing a feature to support students in understanding Snap! concepts, an REU provided her experience working with help libraries saying “even for me, it’s not always easy to find the solution to the problem that I want. And so I feel like [the current design] is not going to be the easiest thing for [the students]”, and then proceeded to suggest changes to the teacher’s design.

The REUs were proactive and provided direct support to teachers in an effort to improve the design. The REUs asked probing questions to help teachers clarify and improve their designs. For example, the teachers often forget to consider the design of the student portal, so REUs frequently asked, “What would this look

like from a students’ perspective?”. The REUs were expected to start implementing the features the teachers created after the completed design sessions. It may be that the responsibility to develop the designs motivated REUs to create a clear and complete design. During the prototyping sessions, the REUs helped the teachers, one-on-one, with their designs and using Figma. The REUs were knowledgeable enough about the software that they could answer questions about how to add the requested functionality. Not only did the teachers share their current practices and design ideas with the REUs, they also shared their domain knowledge. One example was when the teachers clarified the difference between formative and summative assessments. This was helpful in introducing specificity into the design process, clarifying how the experience should work for the teachers and students as well as why these features were needed from a pedagogical perspective.

4.3 Tensions between MY class needs and ALL class needs in design - RQ2

Even though the teachers all taught introductory computing, teachers had different needs based on their teaching styles, environments, and students. In this study, teachers taught different grade levels (US middle grades 6-8 and high school grades 9-12) at different schools. Each school has its own curriculum requirements, assessments, and technologies. Teachers also differed in whether or not their computing lessons were in standalone classes (APCS) or integrated into core classrooms (math or science). We found these differences may have resulted in a less collaborative design process, focusing more on teachers’ individual needs, rather than finding common ground. For example, when one middle school teacher suggested that SnapClass automatically save the students’ work, the high school teacher responded that students should be responsible for saving their own work and that “there are certain things we just have to keep reminding them”. Instead of affirming the needs of one another and incorporating them into the design, ideas that did not fit all needs were sometimes dismissed. When discussing adding peer code reviews to SnapClass, the middle school teachers were excited that students could see each other’s code and add it to their own programs. However, the high school teacher was afraid this may result in plagiarism, which may be flagged by the College Board, a national U.S.-based college course credit certification program. After hearing the high school teacher’s perspective, one middle school teacher suggested the feature was not necessary and could be removed from the list of features. Instead of discussing how to design peer reviews to fit the needs of both classrooms, the teachers chose to end the discussion.

4.4 Teacher Design Disengagement / Pain Points - RQ2

Teachers experienced difficulties when engaging in the PD process. First, teachers struggled to learn and use prototyping software, Figma. One teacher described learning the software as “painful” and not necessary. Additionally, while the teachers were excited to share about their current practices, some teachers seemed less enthusiastic to ideate solutions. One teacher explained that many of her problems could be solved with “classroom management” and shared that many of the design ideas for SnapClass were unnecessary. For example, instead of a “built in reward for somebody who goes above and beyond”, teachers should identify when someone needs recognition. Teachers struggled with and were less likely to consider their students’ perspective in their designs. When asked what the student portal would look like when filling out a self-assessment, one teacher said that it would look exactly like the teacher portal. Details on how the self-assessment would be assigned and accessed by a student were not planned for. Teachers required continuous encouragement and reminders to engage with the physical tools for making, like the paper and the pens during the Future Workshop. One teacher was reluctant to use the tools and stated “I just, I don’t like writing” or “drawing”.

4.5 Programming the Designs - RQ3

After the design sessions, the REU students had 6 weeks to program these features. They decided to only focus on three features that were agreed upon by the teachers. These features were self-assessment assignments, assignment differentiation, and student help request. By the end of the summer experience, the REUs only partially implemented one of the three designs produced by the teachers. While the REUs enjoyed preparing and leading the design sessions, they felt unprepared to implement the designs into SnapClass. REUs sometimes asked to meet with the teachers to clarify the design, but little feedback was given as the features were unfinished. Although the REUs could not complete features due to the time constraints and the scope of the designs, a new team of senior undergraduate computer science majors implemented the designs in one semester. Because the development took place outside of the summer program, the undergraduate developers were not able to receive input directly from the designer teachers.

5 LESSONS LEARNED

5.1 Together in the Physical Space

Teachers required continuous encouragement and reminders to engage with the physical tools for making, like the paper and the pens during the Future Workshop. One solution to this challenge may be to ensure that in a small PD environment like this one, that there be one REU for each teacher. Instead of the teachers sitting across the table from the REU facilitators, they could sit side by side and work together with them. Teachers also had difficulty learning how to use the Figma software during *Making* which led to disengagement in the tasks and requests for continuous support from the REUs. In future implementations, with enough REUs, the students could use Figma while the teachers tell them what to build - similar to a driver/navigator style of pair programming [41]. The

teachers would then not be required to learn the software and the REUs could ask more pointed questions about the design.

5.2 REU Leadership & Facilitator Training

The mentors frequently needed to intervene during the design sessions to clarify tasks and explain the motivation for completing each task. The REUs sometimes asked the mentors how they should proceed during the session. Although REUs practiced the PD sessions with the mentors roleplaying as teachers, it seemed that the mentors had too much prior knowledge about PD to behave like real participants. In future implementations, practice sessions should be run with participants with little to no design experience. Feedback from those groups may be more authentic and help the REUs refine their plans to better explain the task procedure and the motivation for each task.

The REUs were uncertain about when they should probe for more information and when to transition to the next design step. This led the REUs to move through the steps of the process quickly and miss out on pertinent feedback from the teachers. These results echo findings from past research that novice HCI practitioners tend to be less comfortable and confident than professionals who have experience employing HCI methods [1]. When instructing REUs on how to facilitate, it is important to instruct them to let the conversation continue as long as it’s productive and relevant. REUs should embrace the uncomfortability of silence to give teachers time to reflect and share more information. Lastly, as facilitators, REUs should be instructed to be direct with the teachers to ensure productive design sessions. Many activities (such as prototyping first using paper and pen) have been frequently used successfully in other PD sessions. While there is always space for accommodation and improvisation, teacher hesitation cannot fully dictate the direction of the co-design sessions.

Allowing REUs to create PD protocols gave them freedom to take ownership of the sessions. However, this ability to adapt sessions meant they strayed from traditional PD research. Instead of following a predefined Enact method, the REUs chose a less robust method. It may be beneficial to plan the co-design methods in advance, and teach REUs how to modify the methods as needed to accommodate teacher needs or resistance. While the mentors did provide research papers with examples of make, tell, and enact methods, the REUs did not choose predefined methods from all three. To maintain some autonomy and feelings of ownership, mentors could choose the methods, create a template protocol, and assign related reading, but allow REUs to adapt the protocols based on the readings and with mentor approval.

Other institutions with similar undergraduate research programs or courses wanting to provide students with real HCI experience may benefit from our findings. While our results are from a collaboration between REUs and teachers, many of our lessons learned may translate to new design contexts. For example, while our REUs needed a better understanding of assessments, it may be that undergraduate students simply need knowledge of the domain of their participants. Similarly, before students are prepared to lead, they need practice sessions with participants with a similar level of design knowledge as their intended participants and predefined methods to adapt to their context. Using our lessons learned and

the next iteration of this program, we hope to provide guidance for undergraduate computer science organizations who want to design and create software, but do not have access to HCI professionals.

5.3 Teachers' Design and Knowledge Contributions

REUs were unfamiliar with some of the teaching domain vocabulary, like knowing the terms “formative” and “summative” assessment. While REUs were responsible for teaching PD methods, it may help if teachers instructed the REUs on pertinent domain knowledge. In future implementations, teachers could collaborate on a slide deck before the start of the PD sessions. Additionally, each teacher could add an introduction about themselves and their class contexts. In addition to helping the REU facilitators understand the teaching domain better, this may help the teachers feel more comfortable collaborating before the design sessions begin.

Another encountered issue was how teachers sometimes dismissed the needs of other teachers because those needs did not apply to them. To help reduce this, the teachers' introductions could include their school context, attributes of their students, and their teaching values. Although we asked teachers to share these verbally, a visual representation on slides may be more memorable, ensure that everyone shares, and make the context more explicit. This may build empathy between the teachers when they have a better understanding of each other's classrooms. Another way teachers may become more invested in each other's needs would be to instruct the teachers to design for one another. After the “Tell” sessions, the teachers could swap struggles and design for each other before designing for themselves. Finally, we observed that the teachers only designed from the teacher perspective - designing for the teacher portal. The facilitators needed to probe the teachers to elaborate on the student portal. The process could be better scaffolded to ensure all user types are considered in the design. Ideally, we would include high school students as participants in the design process to ensure both perspectives are considered. If this is not possible, teachers could start by designing for their students before transitioning to the teacher portal.

5.4 Scaffolding Programming for the Undergraduates

REUs were unable to program the designed features in time. The features the teachers selected were the most complex of those brainstormed. For example, one teacher designed a hand-raising feature where students requesting help are placed in a queue and peer helpers are selected and alerted. This feature was more complex than, for example, a frequently asked questions page that was not selected. It may be more realistic for teachers to work together and design a single feature. Alternatively, employing other programmers to finish implementation may work. In future iterations, it may make sense to give REUs a week before the PD sessions start to become familiar with the code base. Scaffolding their learning of the code base may include giving them small tasks (e.g. changing colors of buttons). This would ensure REUs are not trying to learn the entirety of the code base while programming their first feature. SnapClass is in a very early stage of development. It may be that with additional structure and further into its development,

there will be more code for REUs to reuse/adapt in turn making the programming tasks faster. Overall, we wished for REUs to receive more feedback from the teachers about their implementation choices. With the lack of programming scaffolding, REUs were unable to do multiple development iterations and receive important feedback from the teachers.

6 CONCLUSION

Educational software is rarely co-designed with teachers and students which means that the realities of the classroom, including teachers' needs and workflows, may be overlooked in software design. We have created a novel co-design process that can help both teachers and students learn about HCI and research methods while contributing to the design of educational systems that will benefit future students and teachers. We mentored two Research Experience for Undergraduate students (REUs), to organize and lead PD sessions with three teachers (of grades 6-12) in the summer program.

Due to resource constraints, it can be difficult for students and research labs, or small development teams, to hire HCI experts, which may result in poorly designed software. Finding ways like our novel PD method to enable inexperienced HCI teams and clients to conduct co-design sessions - although imperfect - may result in better software creation. However, there is little guidance in the literature to inform the process of empowering inexperienced teams to engage in participatory co-design.

Therefore, we investigated the outcomes of the design collaboration and provided recommendations for such future design collaborations. We found that this process exposed teachers and undergraduates to HCI techniques, empowered undergraduates to lead design sessions, and encouraged knowledge transfer between the groups. The development team members (students in our case) need training in the common language of the domain of their participants (teachers in our case), practice sessions with similar participants, and predefined PD methods to adapt to their context.

One limitation of this work was that it was small in nature. It was the researchers' first time leading this PD program and the cohort of undergraduates and teachers was small. Another limitation that we consider is how these findings can be applied to other PD programs. While students/teams may receive similar benefits working with tools and participants from another domain, it's unclear whether participants in another domain would spend a similar amount of time investing in the PD process. In future iterations, we will take the lessons learned to increase student developer confidence and mitigate teacher participant reluctance. Our insights can also provide insights for other teams to empower undergraduate students or software team members newer to HCI to lead participatory design sessions.

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